

ECON 351 - Marginal Utility Guide

Total Utility, Marginal Utility, and the Second Derivative

What this guide covers:

Total Utility (TU) - overall satisfaction from wealth or goods

Marginal Utility (MU) - extra satisfaction from one more unit (1st derivative)

Second derivative - whether MU is rising, falling, or flat

How this connects to risk attitudes on the exam

1. Total Utility (TU)

Total Utility is the overall satisfaction you get from a given amount of wealth or goods.

Think of it as the height of your happiness at a specific quantity.

- $TU(0)$ = satisfaction from zero units
- $TU(3)$ = total satisfaction from 3 units
- TU generally rises as you get more, but not always at the same rate

Example: If 1 slice of pizza gives 10 utils total and 2 slices give 25 utils total,

then $TU(1) = 10$ and $TU(2) = 25$.

2. Marginal Utility (MU) - First Derivative

Marginal Utility is the additional satisfaction from getting exactly ONE more unit.

In calculus terms: $MU = dU/dx$ (or $u'(I)$ for income).

Discrete version (what you often use on homework):

- MU at unit 2 = $TU(2) - TU(1)$
- MU at unit 3 = $TU(3) - TU(2)$
- MU answers: 'How much extra utility do I get from the next unit?'

Example: $TU(1) = 10$ and $TU(2) = 20$

MU at unit 2 = $20 - 10 = 10$ additional utils from the 2nd unit.

3. Second Derivative - Is MU Increasing or Decreasing?

The second derivative measures the rate at which marginal utility is changing.

It tells you whether satisfaction from each extra unit is:

- Accelerating (MU rising) -> second derivative > 0 -> risk-loving
- Slowing down (MU falling) -> second derivative < 0 -> risk-averse
- Staying flat (MU constant) -> second derivative $= 0$ -> risk-neutral

KEY EXAMPLE (increasing marginal utility):

At unit 1, MU = 10 additional utils (going from 0 to 1 gives +10).

At unit 2, MU = 20 additional utils (going from 1 to 2 gives +20).

MU went from 10 to 20 -> marginal utility is INCREASING.

So the second derivative is positive -> risk-loving preferences.

4. Compare: Decreasing vs Increasing Marginal Utility

Most exam utility functions show diminishing marginal utility (risk averse).

- Risk averse: MU falls as you get more (e.g. $MU = 9/I$, or $u = \sqrt{I}$)
- Risk neutral: MU stays constant (e.g. $u = I$, so $MU = 1$ always)
- Risk loving: MU rises as you get more (e.g. $u = I^2$, so $MU = 2I$ rises with I)

Decreasing MU example: $TU(1)=10$, $TU(2)=18 \rightarrow MU \text{ at } 2 = 8$ (less than 10).

Increasing MU example: $TU(1)=10$, $TU(2)=30 \rightarrow MU \text{ at } 2 = 20$ (more than 10).

Exam shortcut: $u''(I) < 0$ risk averse | $u''(I) = 0$ risk neutral | $u''(I) > 0$

risk loving

5. Quick Reference Table

Concept	Meaning	Calculus	Risk attitude
Total Utility	Overall satisfaction	$U(x)$ or $u(I)$	-
Marginal Utility	Extra utility from +1 unit	$u'(I)$ or dU/dx	-
MU decreasing	Each next unit adds less	$u''(I) < 0$	Risk averse
MU constant	Each next unit adds same	$u''(I) = 0$	Risk neutral
MU increasing	Each next unit adds more	$u''(I) > 0$	Risk loving

Remember: the second derivative is about whether MU itself is going up or down.
If MU at 1 is 10 and MU at 2 is 20, MU is increasing -> risk-loving.